

## 0–6h Lead Time

| Model          | RMSE | Bias % | NSE  | KGE  |
|----------------|------|--------|------|------|
| Hydrique_ML    | 93   | -11    | 0.57 | 0.82 |
| Hydrique_Curve | 56   | -5     | 0.84 | 0.84 |
| SIG_CNR        | 53   | -2     | 0.89 | 0.90 |
| OFEV           | 82   | +0.7   | 0.78 | 0.71 |

All models show strong performance at short lead times. NSE values above 0.5 indicate that models explain most of the variance of observed peak flows. SIG\_CNR performs best, with NSE close to 0.9 and very low bias (-2%), meaning it reproduces both magnitude and variability accurately.

## 6–12h Lead Time

| Model          | RMSE | Bias % | NSE   | KGE  |
|----------------|------|--------|-------|------|
| Hydrique_ML    | 200  | -33    | -1.08 | 0.49 |
| Hydrique_Curve | 93   | -12    | 0.54  | 0.75 |
| SIG_CNR        | 99   | -8     | 0.38  | 0.72 |
| OFEV           | 151  | +3.7   | -0.82 | 0.11 |

Hydrique\_ML has  $NSE < 0$ , meaning it performs worse than using the mean observed peak (climatology). This is due to strong underestimation (~ -33%). SIG\_CNR remains usable, as  $NSE > 0$  indicates it still captures part of the variability.

## 12–18h Lead Time

| Model          | RMSE | Bias % | NSE   | KGE   |
|----------------|------|--------|-------|-------|
| Hydrique_ML    | 219  | -30    | -1.43 | 0.33  |
| Hydrique_Curve | 132  | -14    | 0.15  | 0.69  |
| SIG_CNR        | 81   | -6     | 0.47  | 0.74  |
| OFEV           | 158  | +1.6   | -2.28 | -0.09 |

Strong divergence appears between models. Hydrique\_ML and OFEV have strongly negative NSE values, indicating very poor predictive skill. SIG\_CNR remains the most reliable, with  $NSE \sim 0.47$  meaning it still captures peak variability.

## 18–24h Lead Time

| Model | RMSE | Bias % | NSE | KGE |
|-------|------|--------|-----|-----|
|-------|------|--------|-----|-----|

|                |     |      |       |      |
|----------------|-----|------|-------|------|
| Hydrique_ML    | 193 | -27  | -0.91 | 0.44 |
| Hydrique_Curve | 139 | -18  | 0.05  | 0.68 |
| SIG_CNR        | 120 | -15  | 0.48  | 0.72 |
| OFEV           | 123 | -2.7 | 0.43  | 0.69 |

Hydrique\_Curve shows NSE close to 0, meaning it performs similarly to a simple mean estimate. SIG\_CNR remains the most robust model. OFEV shows moderate performance with positive NSE.

## 24–36h Lead Time

| Model          | RMSE | Bias % | NSE   | KGE  |
|----------------|------|--------|-------|------|
| Hydrique_ML    | 242  | -38    | -1.44 | 0.33 |
| Hydrique_Curve | 141  | -20    | 0.20  | 0.73 |
| SIG_CNR        | 148  | -21    | -0.05 | 0.65 |
| OFEV           | 112  | -9     | 0.49  | 0.54 |

Hydrique\_ML has NSE < 0, indicating no predictive usefulness. SIG\_CNR begins to lose skill (NSE slightly negative). OFEV becomes the best model in this window with positive NSE.

## 36–48h Lead Time

| Model          | RMSE | Bias % | NSE   | KGE  |
|----------------|------|--------|-------|------|
| Hydrique_ML    | 248  | -39    | -1.57 | 0.30 |
| Hydrique_Curve | 170  | -27    | -0.13 | 0.68 |
| SIG_CNR        | 154  | -19    | 0.04  | 0.59 |
| OFEV           | 161  | -16    | -0.15 | 0.40 |

All models show NSE values near or below zero, indicating very low predictive skill. Persistent negative bias shows systematic underestimation of flood peaks.

## Overall Conclusion

Model performance decreases significantly with increasing lead time. Short-term forecasts are reliable, while long-term forecasts lose predictive skill. Negative NSE values indicate performance worse than a climatological estimate. SIG\_CNR is the most robust model overall, while Hydrique\_ML shows strong systematic underestimation.